

Breaking Droplet Jumping Energy Conversion Limits with Superhydrophobic Microgrooves

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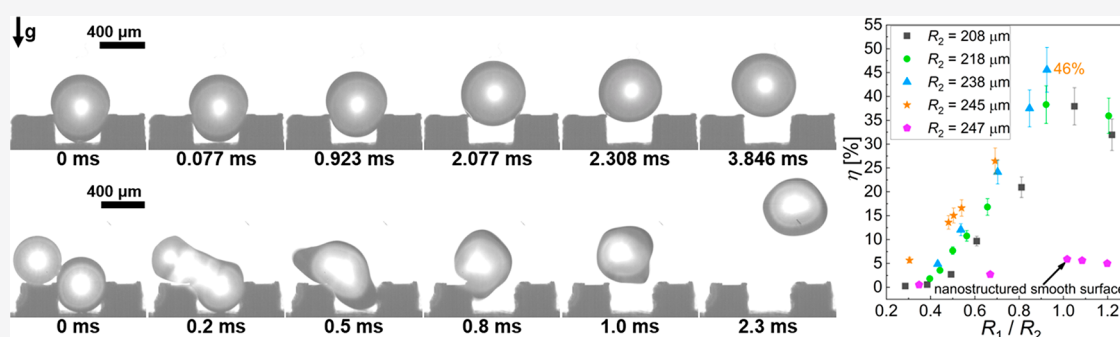
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ABSTRACT: Coalescence-induced droplet jumping has the potential to enhance the performance of a variety of applications including condensation heat transfer, surface self-cleaning, anti-icing, and defrosting to name a few. Here, we study droplet jumping on hierarchical microgrooved and nanostructured smooth superhydrophobic surfaces. We show that the confined microgroove structures play a key role in tailoring droplet coalescence hydrodynamics, which in turn affects the droplet jumping velocity and energy conversion efficiency. We observed self-jumping of individual deformed droplets within microgrooves having maximum surface-to-kinetic energy conversion efficiency of 8%. Furthermore, various coalescence-induced jumping modes were observed on the hierarchical microgrooved superhydrophobic surface. The microgroove structure enabled high droplet jumping velocity ($\approx 0.74U$) and energy conversion efficiency ($\approx 46\%$) by enabling the coalescence of deformed droplets in microgrooves with undeformed droplets on adjacent plateaus. The jumping velocity and energy conversion efficiency enhancements are $1.93\times$ and $6.67\times$ higher than traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces. This work not only demonstrates high droplet jumping velocity and energy conversion efficiency but also demonstrates the key role played by macroscale structures on coalescence hydrodynamics and elucidates a method to further control droplet jumping physics for a plethora of applications.

INTRODUCTION

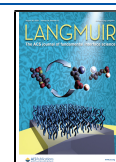
In recent years, nature-inspired functional surfaces, such as lotus leaves,¹ cacti,² and gecko skins,³ have facilitated an increased understanding of superhydrophobicity.^{4–6} Superhydrophobic surfaces show apparent advancing contact angles exceeding 150° with contact angle hysteresis below 10° .⁷ When two droplets coalesce on superhydrophobic surfaces, the released excess surface energy can be partly converted into kinetic energy, resulting in droplet jumping from the symmetry-breaking surface.^{8–10} Superhydrophobic surfaces having a range of structure length scales and surface chemistries have been widely used to improve anti-icing,^{11–15} self-cleaning,^{16–21} and condensation heat transfer performance.^{22–27} To gain a mechanistic understanding of droplet jumping, a plethora of studies have focused on the hydrodynamics of coalescence-induced droplet jumping,^{10,28,29} droplet size effects,^{30–36} jumping velocity,^{9,37–41} energy conversion efficiency,^{42–47} and jumping direction^{48–51} on

superhydrophobic surfaces. These studies have conclusively shown that surface topography, wettability, droplet size, and liquid properties strongly affect coalescence hydrodynamics. From an energy perspective,^{8,30} when two identical-size inviscid droplets coalesce on a surface with negligible surface adhesion, the released excess surface energy is converted into kinetic energy of droplet jumping. The droplet jumping velocity follows an inertial-capillary scaling:⁸ $U \sim (\gamma/\rho R)^{0.5}$, where γ , ρ , and R are the liquid–vapor surface tension, liquid density, and initial radii of coalescing droplets, respectively.

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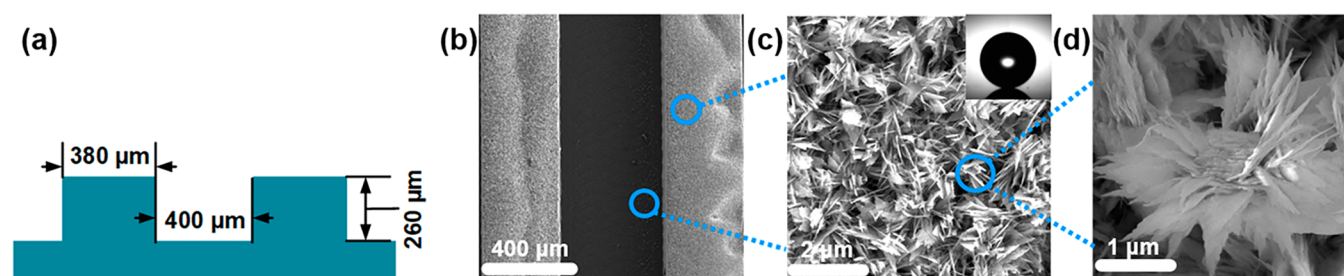


Figure 1. Characteristics of the microgrooved superhydrophobic surface. (a) Schematic of the microgrooved surface with the key dimensions shown. (b) Top-view SEM image of the hierarchical microgrooved superhydrophobic surface. (c) Magnified SEM image of the conformal CuO nanostructures on the hierarchical microgrooved superhydrophobic surface. Inset: optical side-view image of a water droplet residing in the apparent advancing contact angle on the smooth CuO superhydrophobic sample. (d) High-magnification SEM image of the CuO nanostructures showing the bladelike morphology.

Experiments and numerical simulations have depicted the coalescence-induced droplet jumping mechanism as inherently inefficient, having a maximum energy conversion efficiency of $\eta < 6\%$, with jumping velocity $v_j < 0.23U$, due to hydrodynamic momentum transfer limitations during coalescence.^{9,10}

Recently, the use of micro-/macrotextures has provided the possibility to improve coalescence-induced droplet jumping velocity and surface-to-kinetic energy conversion efficiency. Micropillars combined with superhydrophobic nanostructures can realize spontaneous dewetting transition of condensed droplets from valleys to structure tops due to the generated Laplace pressure difference from liquid confinement within the microstructures.⁵² Micropatterned nanowire arrays have been used to improve the condensed droplet mobility and avoid pinning and flooding at high subcooling, in turn achieving high-efficiency jumping droplet condensation.⁵³ In addition, triangular prism structures between coalescing droplets have been used to enable droplet jumping velocities approaching $v_j \approx 0.53U$ with an energy conversion efficiency of $\eta \approx 22.5\%$ by accelerating the retraction of the droplet base area during coalescence.⁵⁴ Inclusion of ridge structures between coalescing droplets has been shown to facilitate coalescence-induced jumping of droplets with low surface tensions and high viscosities, with $\eta \approx 18.8\%$.⁵⁵ Recently, a study demonstrated that coalesced droplets simultaneously impinging onto the bottom and side of the L-shaped macrostructure could increase the energy conversion efficiency by $2.7\times$.⁵⁶ Furthermore, asymmetric V-grooves on superhydrophobic surfaces can result in an enhanced jumping velocity of $v_j \approx 0.61U$ with a $7\times$ increase in energy conversion efficiency compared with droplets jumping on flat superhydrophobic surfaces.⁵⁷ Hence, the introduction of micro-/macrostructures (hierarchy) on superhydrophobic surfaces is clearly a viable pathway to manipulate jumping velocity and direction of coalesced droplets via the control of initial droplet morphology, position, and coalescence hydrodynamics.^{42,58,59}

Here, we explore the role of macrostructures and confinement on coalescence-induced droplet jumping hydrodynamics. Droplet jumping behavior was studied on hierarchical microgrooved and smooth nanostructured superhydrophobic surfaces. While the majority of prior work has focused on droplet jumping using spatially random nucleation, we study well-defined coalescing droplets by deposition of microdroplets generated by a microgoniometer. To enable hydrodynamic manipulation during coalescence, we utilized microgrooves with $400\ \mu\text{m}$ length scales, much larger than the size of the generated microdroplets ($\approx 15\ \mu\text{m}$). Due to the presence of

microgrooves, various droplet jumping modes were observed, including self-jumping of deformed droplets inside microgrooves, coalescence-induced jumping between deformed and undeformed droplets, coalescence-induced jumping of suspended droplets on adjacent microgrooves, and jumping of droplets coalescing on adjacent plateaus. The jumping velocity, direction, and energy conversion efficiency of each jumping mode were quantified, demonstrating the high jumping velocity ($v_j \approx 0.74U$) and surface-to-kinetic energy conversion efficiency ($\eta \approx 46\%$) for coalescence between deformed and undeformed droplets.

EXPERIMENTAL METHODS

Sample Preparation. Square mirror-finished Cu plates (99.9% purity) having thickness and width of 2 ± 0.2 and 28 ± 0.2 mm, respectively, were utilized to fabricate the hierarchical microgrooved and nanostructured smooth superhydrophobic surfaces. Rectangular microgrooves with $260\ \mu\text{m}$ depth and $400\ \mu\text{m}$ width with intervals of $380\ \mu\text{m}$ were manufactured by mechanical milling on the smooth Cu plates, as schematically shown in Figure 1a. Figure 1b shows top-view scanning electron microscopy (SEM, S-4800 Hitachi) images of the hierarchical microgrooved superhydrophobic surface. Conformal nanoscale structures on the microgrooved surface were fabricated through CuO nanostructuring of the Cu samples (Figure 1c,d). Sharp bladelike CuO nanostructures were synthesized by a self-limiting chemical oxidation reaction.^{60,61} First, in order to remove the organic contaminants, the smooth and microgrooved Cu plates were ultrasonically cleaned for 10 min with acetone, isopropanol, and ethanol, successively. The samples were then rinsed by deionized (DI) water and dried in a clean nitrogen (N_2) gas stream. The cleaned samples were immersed into a $2.0\ \text{M}$ hydrochloric acid solution for 2 min to eliminate the original oxide layer and then rinsed with DI water and dried within a clean N_2 stream. The samples were then immersed into an alkaline solution composed of NaClO_2 , NaOH , $\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$, and DI water ($3.75:5:10:100$ wt %) at $96 \pm 4\ ^\circ\text{C}$ for 15 min to synthesize the CuO nanostructures. Then, the samples were rinsed by DI water and dried within a clean N_2 stream. Finally, heptadecafluorodecyl trimethoxysilane was deposited on the samples to achieve hydrophobic functionalization through a vapor phase deposition method. The samples were placed in a sealed container with a vial of $1\ \text{mL}$ of HTMS toluene solution ($5:95$ vol %) and then heated in an atmospheric pressure oven at $85\ ^\circ\text{C}$ for 3 h.⁶² This process allowed for the development of a conformal self-assembled monolayer of HTMS on the samples.

The CuO nanostructures had sharp blade shapes with heights of $\sim 1\ \mu\text{m}$, blade thicknesses of $< 100\ \text{nm}$, and average blade widths of $\sim 300\ \text{nm}$, with a tip dimension of $\sim 10\ \text{nm}$. The functionalized CuO nanostructures showed water droplet apparent advancing contact angles of $168.6 \pm 2.7^\circ$ with a contact angle hysteresis angle of $5.1 \pm 3.2^\circ$ as measured by a microgoniometer (MCA-3, Kyowa Interface Science), and shown in Figure 1c, inset.

Experimental Processes. To observe droplet coalescence and measure jumping velocity and direction, a visualization experimental platform combining high-speed and high-magnification optical imaging with a frequency-controlled microdroplet dispenser was constructed as shown in Figure 2. The experiment used a high-speed

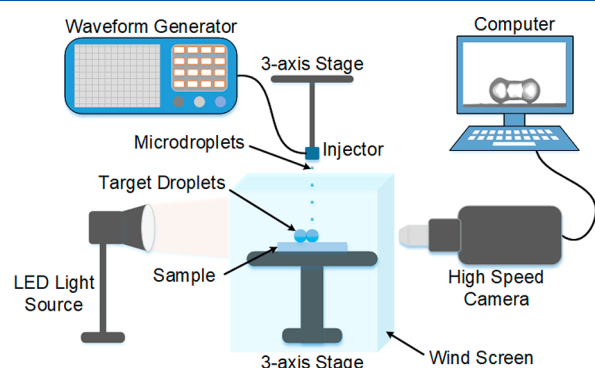


Figure 2. Schematic of the experimental apparatus. Monodisperse microdroplets with $\approx 15 \mu\text{m}$ radii produced by the piezoelectric pulse injector deposit on the sample to form a larger target droplet. The size of the target droplet is adjusted by balancing the amount of injecting microdroplets and the liquid loss due to evaporation, while the printing location of the target droplet is controlled by the motion of independent three-axis stages supporting the sample and injector.

camera (Phantom v711, Vision Research) coupled to a magnification lens (TU Plan Fluor, Nikon), a piezoelectric pulse injector integrated in a microgoniometer (MCA-3, Kyowa Interface Science), and an

illumination subsystem (TSPA22x8, AITECSYSTEM). The piezoelectric injector was installed above samples with an adjustable spacing ranging from 3 to 10 mm. The injector would produce monodisperse microdroplets with radii of $\approx 15 \mu\text{m}$ deposited on the test surface to form larger target droplets for later coalescence. The velocity and frequency of the monodisperse microdroplets were regulated by the working voltage and input frequency of a waveform generator. The higher the working voltage, the larger velocity of monodisperse microdroplets. The input frequency determined the number of injecting microdroplets per second. Depending on the size of target droplets in our experiments, the working voltage and frequency of the waveform generator were adjusted to be in the ranges 6.6–7.2 V and 100–500 Hz, respectively. Under these operating conditions, the injecting monodisperse microdroplet approached the target droplet with a velocity of $\sim 0.1 \text{ m/s}$, which led to a 1% deviation of droplet jumping velocity when the initial radius of the target droplet was over $200 \mu\text{m}$.³⁹ Here, the initial radii of all target droplets were larger than $200 \mu\text{m}$. Furthermore, the kinetic energy of the injecting microdroplet ($\sim 7 \times 10^{-14} \text{ J}$) was approximately 5 orders of magnitude smaller than the minimal released surface energy for droplet jumping in our study ($\sim 10^{-9} \text{ J}$). Thus, the effect of injecting monodisperse microdroplets on droplet jumping was negligible.

Differing from the previous studies of coalescence-induced droplet jumping, in which droplet coalescence is triggered by condensate growth,^{59,63} Leidenfrost effect,^{64,65} air blowing,³⁸ or wire pushing,⁵⁴ coalescence was initiated by a gradually growing droplet by the deposition of microdroplets. Our method reduces the impact of undesired external disturbances and energy input prior to coalescence and can control the printing location of initial target droplets through adjusting the independent three-axis stages supporting sample and injector. At first, we used a high-frequency setting on the waveform generator (400–500 Hz) to make an initial target droplet grow

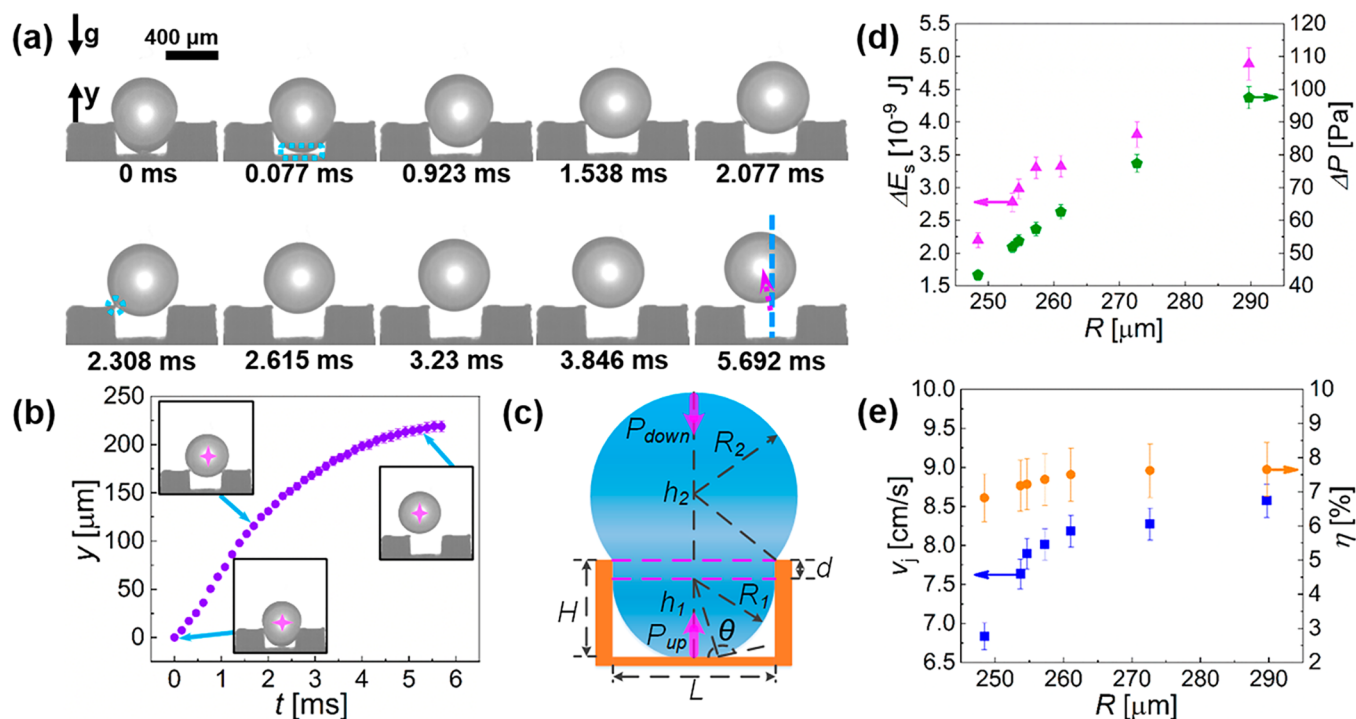


Figure 3. Self-jumping of deformed droplets from microgrooves. (a) High-speed time-lapse images of self-jumping of a deformed water droplet with an equivalent radius, $R = 273 \pm 4 \mu\text{m}$ (Video S1). The equivalent radius is the radius of a spherical droplet having identical volume with the deformed droplet in the microgroove, which is calculated by the assumed model of deformed droplet (Section S2, Supporting Information). The deformed droplet detaching from the valley of the microgroove is circled in the blue dotted frame. The blue dotted line and purple dotted arrow show that the droplet jumping deviates from the surface normal. (b) Jumping trajectory of the deformed droplet. The symbol y represents the jumping height of the center of mass of the droplet. (c) Schematic of a deformed droplet residing in a microgroove. (d) Laplace pressure difference (ΔP) and released surface energy (ΔE_s) of deformed droplets having different equivalent radii. (e) Experimentally measured self-jumping velocity (v_j) and surface-to-kinetic energy conversion efficiency (η) of deformed droplets having different equivalent radii.

rapidly to the desired size. We then adjusted the three-axis stages to shift the printing location so a secondary droplet was formed near the initial target droplet with a prescribed spacing. The spacing between the secondary and primary droplet was rationally controlled such that coalescence radii could be controlled. Once the secondary droplet grew to the desired size, we reduced the frequency of the waveform generator (100–200 Hz) and shifted continually the printing location between two droplets to maintain the liquid lost due to evaporation and maintain the size of the secondary droplet until the two droplets coalesced.

The process of coalescence-induced jumping was recorded using a high-speed camera with high temporal and spatial resolution. Imaging was done with a capture frequency of 10 000–13 000 frames per second and a magnification of 30 $\times$ using a custom lens. The illumination subsystem consisted of a light emitting diode (LED) light source, where the light intensity could be adjusted to ensure a clear image. A transparent cage was used as a wind screen to eliminate the effect of external airflow on microdroplet injection and coalescence-induced droplet jumping. Due to the LED light source and camera being fixed in our microgoniometer setup, we could only observe the experimental phenomenon from the side-view. All experiments were performed at an ambient environment temperature $T_a = 25 \pm 1$ °C (OM-73, Omega) with a relative humidity $\phi = 50 \pm 5\%$ (OM-73, Omega). The droplet jumping velocity and direction were determined according to the trajectory of the jumping droplet captured by the high-speed camera, the details of which are reported in a previous study.³⁹

RESULTS AND DISCUSSION

Droplet Self-Jumping in Microgrooves. To study the effect of the confining microgroove structures on droplet jumping, experiments were conducted on the hierarchical microgrooved superhydrophobic surface. Our ability to control the microdroplets' deposition location enabled us to induce droplet growth inside the valley of a single microgroove. When the growing droplet reached a comparable size scale to that of the microgroove width, the droplet stretched due to confinement from the superhydrophobic side walls. As the droplet grew further, droplet self-jumping from the microgroove occurred (Figure 3a). From a force analysis, the liquid–vapor radius of curvature of the lower part of the deformed droplet (in the groove valley) is smaller than the upper part emerging from the microgroove. The curvature difference results in a Laplace pressure difference, $\Delta P = \gamma(1/R_1 - 1/R_2)$, in the deformed droplet. Here, γ represents the liquid–vapor surface tension of water (≈ 72 mN/m), and R_1 and R_2 are the curvature radii of the lower and upper parts of deformed droplet residing in the microgroove, respectively, as shown schematically in Figure 3c. When the force generated by the Laplace pressure difference exceeds the surface adhesion force, the deformed droplet can detach from the valley of microstructures and spontaneously move upward.^{52,53} In terms of energy arguments, the surface area of the deformed droplet is larger than the spherical droplet with identical volume, which results in surface energy storage due to deformation of the droplet caused by confinement inside the microgroove. During droplet recovery to a spherical shape, the released excess surface energy is partly converted into kinetic energy for droplet jumping. As a result of the slight difference of droplet pinning at the top edges of the microgroove, jumping can slightly deviate from the normal direction (Figure 3a). It should be noted that previous studies employing microgrooves have not observed droplet self-jumping during water vapor condensation.^{66,67} The lack of self-jumping was most likely due to the random nature of condensate nucleation,

resulting in coalescence within the groove between droplets at valleys and groove tops. Furthermore, droplet nucleation sizes were much smaller than the characteristic size of micro-/nanostructures,^{53,68} resulting in a lack of coherent single droplet formation with high-fidelity confinement and high surface adhesion due to the presence of partially wetting droplet growth during condensation. Although self-jumping was not observed before, deformed droplet motion in microgrooves due to Laplace pressure differences during condensation has been reported, forming the basis of hierarchical condensation.^{67,69,70}

In order to quantify the jumping velocity, the droplet jumping trajectory was obtained through tracking of the center of mass (Figure 3b). The initial jumping velocity, v_j , was acquired by taking the derivative of the jumping trajectory at the detaching point from surface, $v_j = |\sin \theta_j|^{-1} (dy/dt)_{y=0}$, where θ_j is the angle between the droplet jumping direction and horizontal plane at the detaching point ($t = 0.077$ ms, Figure 3a). To obtain the released surface energy of the deformed droplet to calculate the energy conversion efficiency, a model was developed to calculate the surface areas and volumes of deformed droplets in the microgrooves (Figure 3c). The deformed droplet is assumed to consist of a large spherical cap residing above the microgroove, with a smaller spherical cap inside the microgroove connected with a cylindrical segment between both spherical caps. The computed volumes of the deformed droplets in the microgrooves agreed well with the actual volumes of spherical droplets after jumping with differences less than 4% (Section S2, Supporting Information). The energy conversion efficiency was defined as the ratio of initial kinetic energy of jumping droplet to the released surface energy (Section S3, Supporting Information). Figure 3d,e shows the Laplace pressure difference, released surface energy, jumping velocity, and energy conversion efficiency of deformed droplets having different equivalent radii. The equivalent radius is the radius of a spherical droplet having identical volume with the deformed droplet in the microgroove, which is determined by the model of a deformed droplet (Section S2, Supporting Information). Based on the characteristics of the microgroove structures, the smallest equivalent radius of the deformed droplet enabling self-jumping is approximately 232 μm on our surface, calculated by a model developed in a previous study.⁵⁸ However, because the regions and density of nanostructures likely resulted in some inhomogeneity on the surface, and the utilized impinging technology may also incur different surface adhesion of initial droplets and more penetration into the structures with higher adhesion, the equivalent radii of deformed droplets enabling self-jumping ranged between radii of 248 to 290 μm , which were slightly larger than the threshold value. The increased surface adhesion force requires a larger Laplace pressure difference for self-jumping of the deformed droplet, representing larger degrees of droplet deformation and equivalent radii, which resulted in higher surface energy release. As the equivalent radii of deformed droplets increased from 248 to 290 μm , the generated Laplace pressure difference in the deformed droplets improved by $\sim 1.5\times$, as shown in Figure 3d. The increased Laplace pressure difference at larger equivalent radii resulted in larger momentum generation after depinning. However, the mass of deformed droplets only increased by $\sim 60\%$ with the increase of deformed droplet radii. Thus, we rationalize that the increase in mass of deformed droplets is inadequate for the larger improvement in momentum caused by the increased

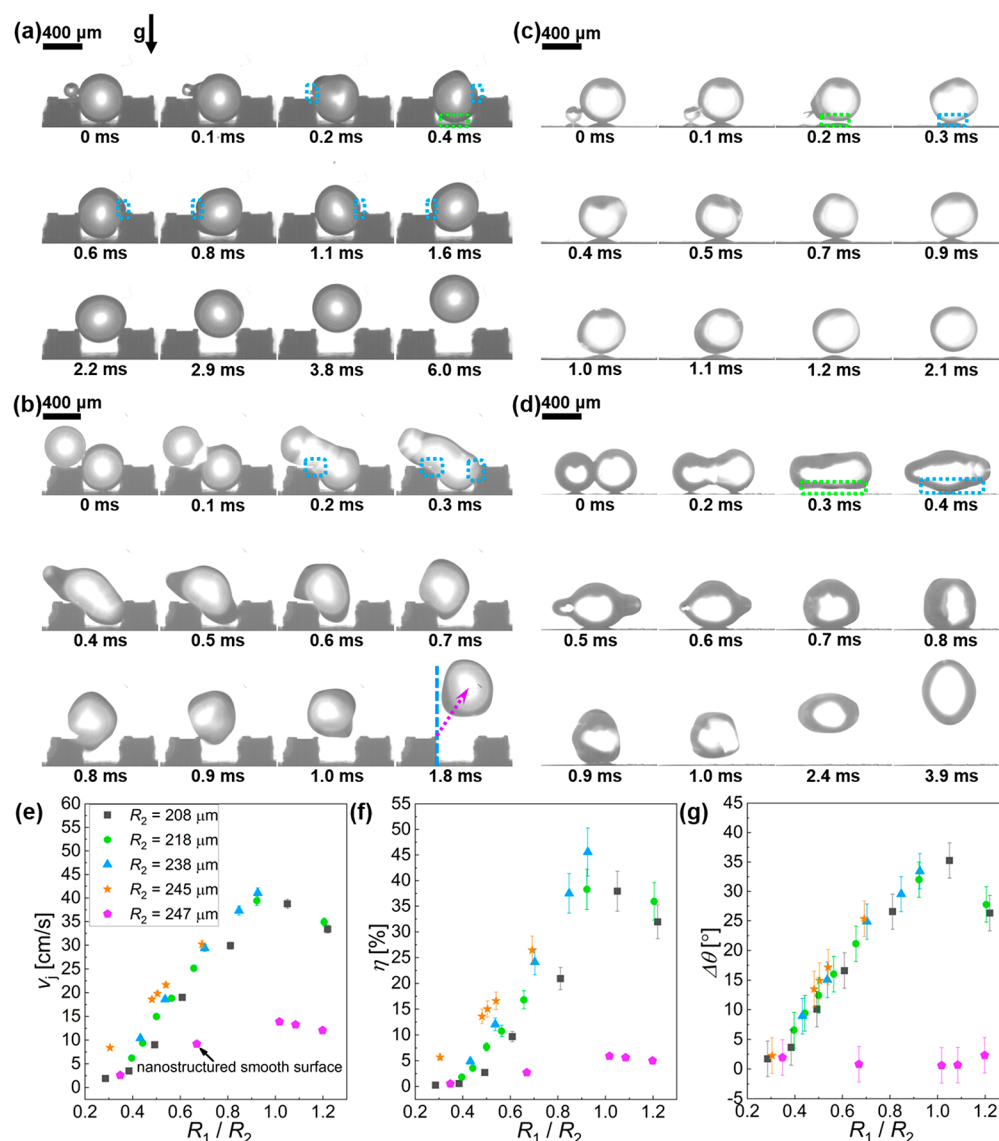


Figure 4. Coalescence-induced droplet jumping on hierarchical microgrooved and smooth superhydrophobic surfaces. (a) Time-lapse images of coalescence-induced droplet jumping of a deformed droplet in a microgroove and a smaller undeformed droplet on an adjacent plateau with $R_1/R_2 = 0.3 \pm 0.01$ ($R_1 = 74 \pm 2 \mu\text{m}$, $R_2 = 245 \pm 4 \mu\text{m}$, Video S2). The coalesced droplet impinging onto both edges of the microgroove is marked with blue dotted frames, and the coalesced droplet detaching from the valley of the microgroove is circled in a green dotted frame. (b) Time-lapse images of coalescence-induced droplet jumping of a deformed droplet in a microgroove and a large undeformed droplet on an adjacent plateau with $R_1/R_2 = 0.93 \pm 0.03$ ($R_1 = 221 \pm 3 \mu\text{m}$, $R_2 = 238 \pm 4 \mu\text{m}$, Video S3). The coalesced droplet impinging onto the edge and side wall of a microgroove is circled in a blue dotted frame; the blue dotted line and purple dotted arrow show that the droplet jumping deviates from the surface normal. (c) Time-lapse images of traditional coalescence-induced droplet jumping on a smooth superhydrophobic CuO surface with $R_1/R_2 = 0.35 \pm 0.01$ ($R_1 = 84 \pm 2 \mu\text{m}$, $R_2 = 242 \pm 4 \mu\text{m}$, Video S4). (d) Time-lapse images of traditional coalescence-induced droplet jumping on a smooth superhydrophobic CuO surface with $R_1/R_2 = 0.91 \pm 0.03$ ($R_1 = 214 \pm 3 \mu\text{m}$, $R_2 = 236 \pm 4 \mu\text{m}$, Video S5). In parts c and d, the initial separation of coalescing droplets with the surfaces is circled in green dotted frames, and the liquid bridges impinging onto the surfaces are marked with blue dotted frames. (e) Droplet jumping velocity v_j as a function of R_1/R_2 . (f) Surface-to-kinetic energy conversion efficiency η as a function of R_1/R_2 . (g) Deviation angle of droplet jumping ($\Delta\theta$) as a function of R_1/R_2 . For clarity, the data point legend for curves in parts f and g is shown in part e.

Laplace pressure difference, in turn resulting in higher self-jumping velocity. Although the kinetic energy of deformed droplets increased at larger equivalent radii, the energy conversion efficiency of self-jumping only slightly improved (roughly constant), due to the simultaneously increasing released surface energy with the increased equivalent radii. Figure 3e shows that the self-jumping velocity and energy conversion efficiency slightly increase with the equivalent radii of deformed droplets. The maximum self-jumping velocity and energy conversion efficiency approached $\approx 8.6 \text{ cm/s}$ and 7.7% ,

when $R = 290 \pm 4 \mu\text{m}$. The energy conversion efficiency of self-jumping was slightly larger than that of coalescence-induced jumping of two droplets with an identical radius on a smooth superhydrophobic surface ($\approx 5.6\%$ ³⁹). The self-jumping results indicated that the abilities to confine the droplet growth and control the depinning in a favorable momentum transfer plane were key factors for enhancing energy conversion efficiency. Hence, we investigated whether it is possible to further improve velocity and energy conversion efficiency by optimizing the droplet coalescence location.

Coalescence-Induced Jumping between Deformed and Undeformed Droplets.

In addition to self-jumping, we observed coalescence-induced jumping between deformed and undeformed droplets. Figure 4a–d shows time-lapse images of this jumping mode along with traditional coalescence-induced droplet jumping on a smooth superhydrophobic surface. As shown in Figure 4a, the deformed droplet inside the microgroove having an equivalent radius of $R_2 = 245 \pm 4 \mu\text{m}$ coalesced with a smaller undeformed droplet with radius $R_1 = 74 \pm 2 \mu\text{m}$ on the adjacent plateau, resulting in jumping with a radius ratio, $R_1/R_2 = 0.3 \pm 0.01$. During coalescence, the small undeformed droplet on the plateau was sucked into the deformed droplet in the microgroove. Due to the suppression of lateral expansion of the coalesced droplet by the sidewalls of the microgroove, the liquid–gas interface of the oscillating droplet would impinge repeatedly on both edges of the microgroove to provide a driving force for droplet depinning and jumping. Once the coalesced droplet detached from the valley of the microgroove at $t = 0.4 \text{ ms}$, it jumped normal to the surface due to the liquid–gas interface impinging on both edges of the microgroove. The jumping behavior was very different from traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces having droplets with a similar radius ratio. Figure 4c shows two droplets with a radius ratio of $R_1/R_2 = 0.35 \pm 0.01$ coalescing and jumping on the smooth CuO nanostructured superhydrophobic surface. The coalesced droplet initially detached from the surface at $t = 0.2 \text{ ms}$, followed by liquid-bridge impinging on the surface to initiate droplet jumping in the normal direction. For $R_1/R_2 = 0.3 \pm 0.01$, the coalescence-induced jumping velocity between deformed and undeformed droplets was $v_j = 8.4 \text{ cm/s}$, larger than the traditional coalescence-induced droplet jumping on a smooth superhydrophobic surface shown in Figure 4c ($v_j = 2.6 \text{ cm/s}$), due to the additional Laplace pressure difference and higher released surface energy of the deformed droplet.

As R_1/R_2 increased, the interaction between the coalesced droplet and microgroove structure changed. Figure 4b shows time-lapse images of coalescence-induced jumping of a deformed droplet inside a microgroove and a large undeformed droplet on an adjacent plateau with $R_1/R_2 = 0.93 \pm 0.03$, where $R_2 = 238 \pm 4 \mu\text{m}$ and $R_1 = 221 \pm 3 \mu\text{m}$. During coalescence, a liquid bridge formed between both droplets at $t = 0.1 \text{ ms}$ and began to impinge on the edge of the microgroove to generate a driving force for droplet jumping at $t = 0.2 \text{ ms}$. For traditional coalescence-induced droplet jumping on a smooth superhydrophobic surface having a similar R_1/R_2 , the coalesced droplet initially detached from the surface with a small distance during formation of a liquid bridge and then began to impinge on the surface at $t = 0.3\text{--}0.4 \text{ ms}$ (Figure 4d). However, the liquid bridge did not detach from the surface during the coalescence between deformed and undeformed droplets, since the increased height difference between the centroids of the two droplets created by the plateau changed the formation position of the liquid bridge. Furthermore, due to the suppression of lateral expansion of the coalesced droplet by the sidewalls of the microgroove, the coalesced droplet could impinge on the sidewall of the microgroove to generate more recoil momentum for droplet jumping, as shown in Figure 4b. As a result of the liquid bridge impinging on the edge of the microgroove, a momentum component parallel to the surface plane was generated. The coalesced droplet jumped in a direction deviating from surface normal with an angle of $33 \pm 3^\circ$. The observed phenomenon

explains the previously observed jumping-induced sweeping during vapor condensation on hierarchical microgrooved superhydrophobic surfaces.^{66,67} For such a large radius ratio ($R_1/R_2 = 0.93 \pm 0.03$), the coalescence-induced jumping velocity between deformed and undeformed droplets approached $v_j = 41 \text{ cm/s}$, $1.93\times$ higher than a traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces ($v_j = 14 \text{ cm/s}$).

Figure 4e–g shows the coalescence-induced jumping velocity, energy conversion efficiency, and jumping angle of deformed droplets having different equivalent radii (R_2) as a function of radius ratio (R_1/R_2), respectively. For all sizes of deformed droplets, the coalescence-induced jumping velocity and energy conversion efficiency increased as R_1/R_2 approached 1, consistent with traditional coalescence-induced jumping on smooth superhydrophobic surfaces.^{30,33,39} When the size difference between the two droplets reduces, more surface energy is available for conversion to kinetic energy for droplet jumping. The droplet jumping velocity and energy conversion efficiency were slightly higher for larger equivalent radii of deformed droplets (R_2). This can be attributed to the larger deformed droplet possessing larger Laplace pressure difference and additional release of surface energy after coalescence. When $R_2 > 238 \mu\text{m}$, $R_1/R_2 > 1$ could not be achieved due to the limited space available on the adjacent plateau to make the droplet smaller than the deformed droplet in the microgroove prior to contacting each other.

The groove structures on the surface could intervene at early time scales during coalescence when surface energy release began to initiate and prior to significant oscillatory motion of coalesced droplets, as shown in Figure 4b. Early interaction resulted in effective conversion of released surface energy into kinetic energy for droplet jumping.⁷¹ The coalesced droplet impinging on the edge of the microgroove was similar to droplet impinging on protruding ridges,^{55,57} where momentum during coalescence could be effectively redirected with the help of structures to improve jumping velocity and energy conversion. In addition to the ridges studied here, large Laplace pressure differences and additional surface energy stored in the deformed droplets and the suppression of lateral expansion of coalesced droplet due to confinement of microgrooves enabled a further enhancement in the coalesced droplet jumping velocity and energy conversion efficiency. Thus, high droplet jumping velocity and energy conversion efficiency could be achieved, with $v_j = 41 \text{ cm/s}$ ($\approx 0.74U$) and $\eta \approx 46\%$, when $R_2 = 238 \pm 4 \mu\text{m}$ with $R_1/R_2 = 0.93 \pm 0.03$. The jumping velocity is $\approx 1.16\times$ higher than coalescence-induced droplet jumping with $R = 710 \mu\text{m}$ on the superhydrophobic surface with an asymmetric V-groove ($\approx 19 \text{ cm/s}$).⁵⁷ This can be attributed to the smaller droplet radius and additional surface energy stored in the deformed droplet. The energy conversion efficiency is $\approx 6.67\times$ higher than traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces, where $\eta \approx 6\%$.

Figure 4g shows the deviation angle of droplet jumping from the surface normal ($\Delta\theta$). For traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces, the jumping direction of a coalesced droplet is always perpendicular to the surface with small deviation due to random pinning ($<5^\circ$).^{39,48} The deviation angle of coalescence-induced jumping between deformed and undeformed droplets on our microgrooved surface significantly increased as R_1/R_2 approached unity due to the generated in-plane momentum

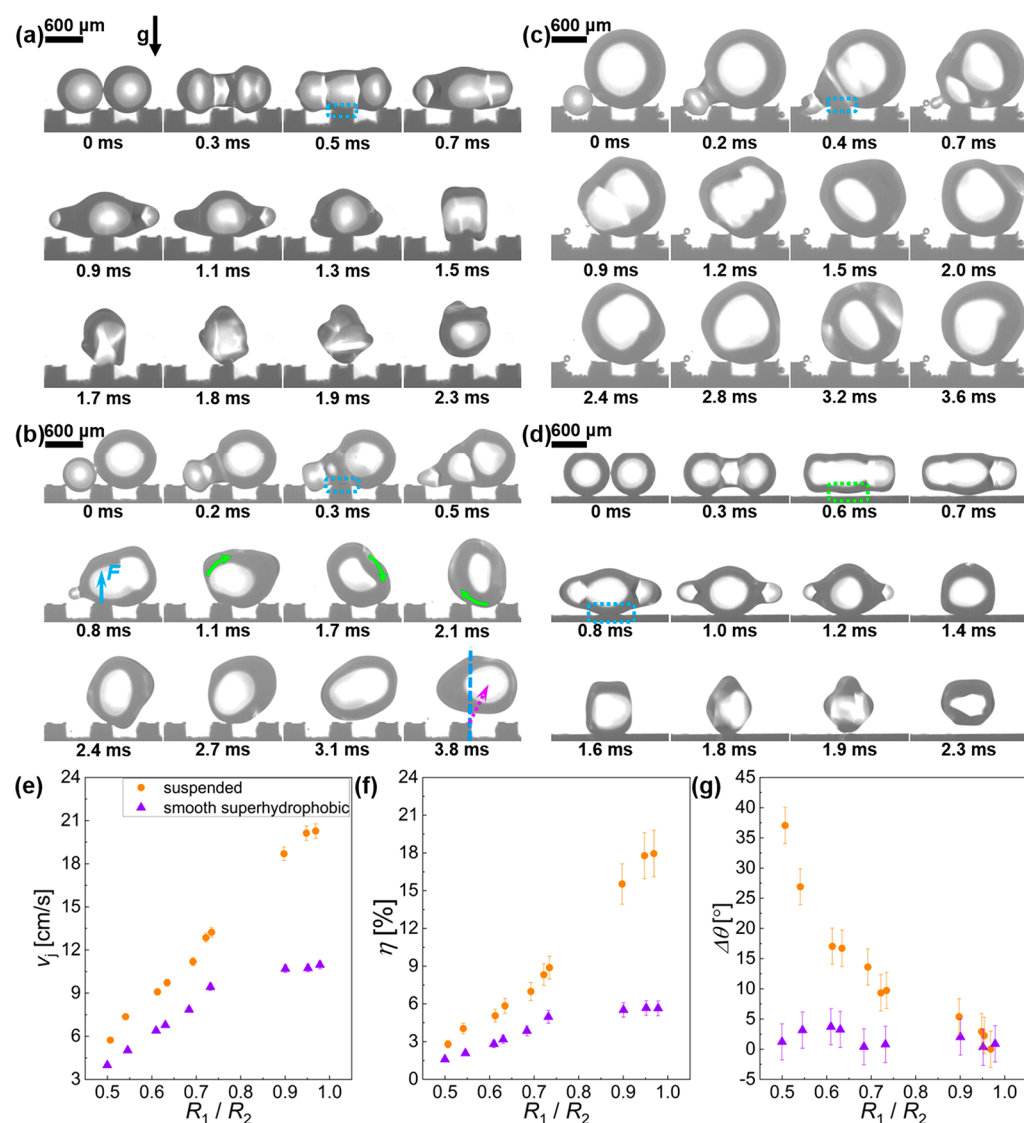


Figure 5. Coalescence-induced jumping of suspended droplets. (a) Time-lapse images of coalescence-induced jumping of suspended droplets with $R_1/R_2 = 0.97 \pm 0.03$ ($R_1 = 382 \pm 5 \mu\text{m}$, $R_2 = 395 \pm 5 \mu\text{m}$, Video S6). (b) Time-lapse images of rotational jumping of suspended droplets with $R_1/R_2 = 0.54 \pm 0.02$ ($R_1 = 285 \pm 4 \mu\text{m}$, $R_2 = 527 \pm 6 \mu\text{m}$, Video S7). The surface counteracting force F , droplet rotation, jumping direction, and surface normal are marked with a blue upward arrow, green curved arrows, a purple dotted line, and a blue dotted line, respectively. (c) Time-lapse images showing no jumping when suspended droplets coalesced with $R_1/R_2 = 0.36 \pm 0.01$ ($R_1 = 251 \pm 4 \mu\text{m}$, $R_2 = 691 \pm 8 \mu\text{m}$, Video S8). (d) Time-lapse images of traditional coalescence-induced droplet jumping on a smooth superhydrophobic CuO surface with $R_1/R_2 = 0.98 \pm 0.03$ ($R_1 = 381 \pm 5 \mu\text{m}$, $R_2 = 389 \pm 5 \mu\text{m}$, Video S9). The initial separation of the coalescing droplet with the surface is marked with the green dotted frame. In parts a–d, the liquid bridges impinging onto the surfaces are marked with blue dotted frames. (e) Droplet jumping velocity (v_j) as a function of R_1/R_2 . (f) Surface-to-kinetic energy conversion efficiency (η) as a function of R_1/R_2 . (g) Deviation angle of droplet jumping ($\Delta\theta$) as a function of R_1/R_2 . For clarity, the data point legend for curves in parts f and g is shown in part e.

caused by the liquid bridge impinging on the edge of the microgroove. The maximum deviation angle was $\Delta\theta = 35 \pm 3^\circ$, presenting a feasible path to regulate droplet jumping direction by rationally designing microgroove structures.

Coalescence-Induced Jumping of Suspended Droplets. In some cases, when the surface adhesion was large enough to prevent self-jumping of deformed droplets in microgrooves, deformed droplets would transition to the tops of microgrooves due to the Laplace pressure difference. As a result of the constant horizontal distance between the centroids of the two suspended droplets ($\approx 780 \pm 10 \mu\text{m}$) on the adjacent microgrooves, once R_1/R_2 was determined, the radii of the two suspended droplets were fixed. Figure 5a–d shows time-lapse images of coalescence-induced jumping of

suspended droplets on adjacent microgrooves and droplets on smooth superhydrophobic surfaces. As shown in Figure 5a, two droplets having $R_1/R_2 = 0.97 \pm 0.03$ coalesced to jump away from the surface. During coalescence, a liquid bridge formed above the ridge plateau between the microgrooves, perpendicularly impinging on top of the ridge plateau, resulting in coalesced droplet detachment from the edges of the two microgrooves. Differing from coalescence-induced droplet jumping on smooth superhydrophobic surfaces (Figure 5d), the resulting droplet did not initially detach from the top of the plateau during coalescence. Due to the protruding plateau between droplets, the in-plane momentum is redirected to out-of-plane momentum during the contraction process of the coalesced droplet.⁵⁵ The redirection of momentum facilitates

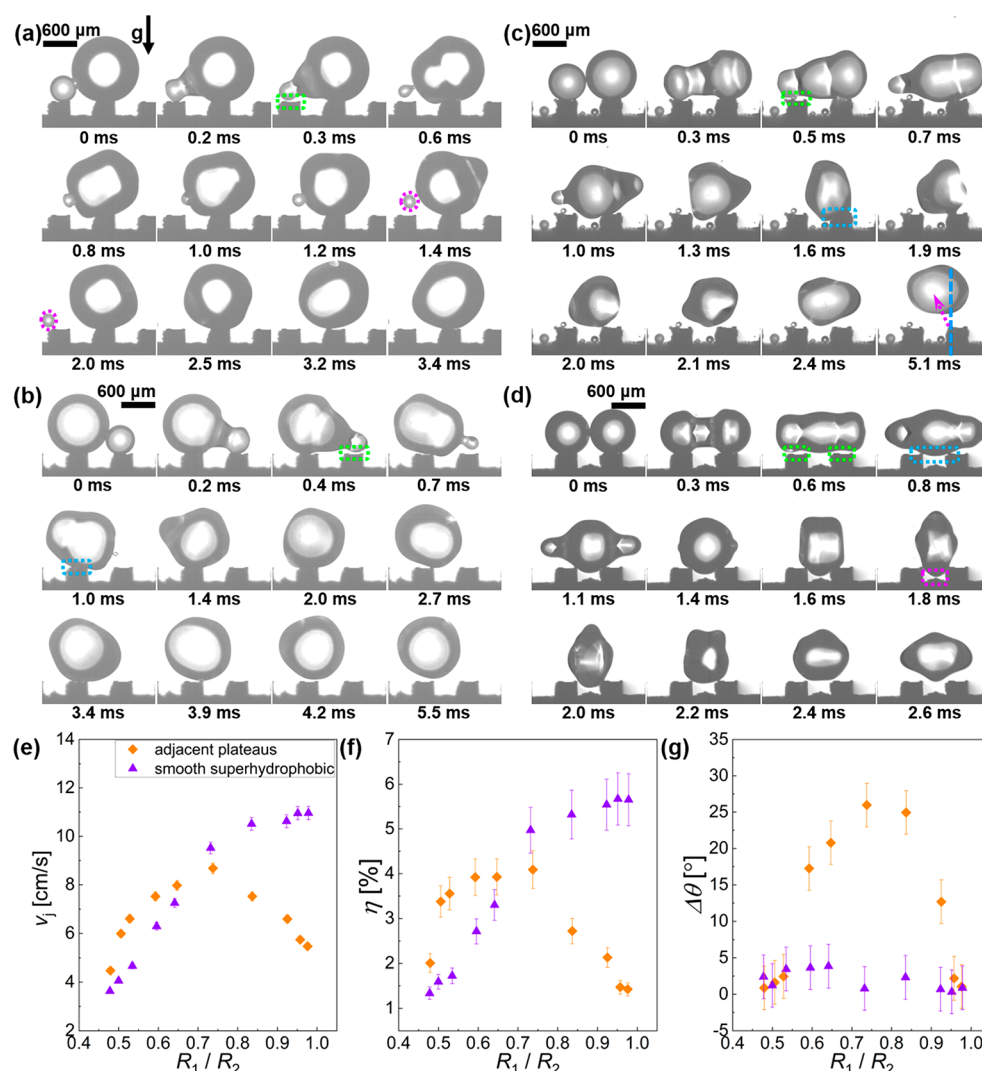


Figure 6. Coalescence-induced jumping of droplets on adjacent ridge plateaus. (a) Time-lapse images showing nonjumping with satellite droplet jetting (dotted purple line) when droplets on adjacent ridge plateaus coalesced with $R_1/R_2 = 0.39 \pm 0.01$ ($R_1 = 232 \pm 3 \mu\text{m}$, $R_2 = 588 \pm 7 \mu\text{m}$, Video S10). (b) Time-lapse images of coalescence-induced jumping of droplets on adjacent plateaus with $R_1/R_2 = 0.48 \pm 0.02$ ($R_1 = 257 \pm 4 \mu\text{m}$, $R_2 = 536 \pm 6 \mu\text{m}$, Video S11). (c) Time-lapse images of coalescence-induced jumping of droplets on adjacent plateaus with $R_1/R_2 = 0.74 \pm 0.03$ ($R_1 = 328 \pm 4 \mu\text{m}$, $R_2 = 445 \pm 5 \mu\text{m}$, Video S12). The jumping direction and surface normal are marked with a purple dotted arrow and blue dotted line, respectively. (d) Time-lapse images of coalescence-induced jumping of droplets on adjacent plateaus with $R_1/R_2 = 0.98 \pm 0.03$ ($R_1 = 384 \pm 5 \mu\text{m}$, $R_2 = 392 \pm 5 \mu\text{m}$, Video S13). The initial separation of coalescing droplets is demarked with the green dotted frame; the impinging of the coalesced droplet onto both edges of the microgroove is demarked with the blue dotted frame, and the droplet nonimpingement with the bottom of the microgroove is demarked with the purple dotted frame. In parts a–c, smaller droplets detaching from ridge plateaus are demarked with green dotted frames, and the coalesced droplets impinging onto the ridge plateaus are demarked with blue dotted frames. (e) Droplet jumping velocity (v_j) as a function of R_1/R_2 . (f) Surface-to-kinetic energy conversion efficiency (η) as a function of R_1/R_2 . (g) Deviation angle ($\Delta\theta$) of droplet jumping as a function of R_1/R_2 . For clarity, the data point legend for curves in parts f and g is shown in part e.

improved energy conversion efficiency. Meanwhile, suspended droplets have smaller surface adhesion due to reduced contact areas with the surface. The coalesced droplet detached from the top of the plateau and jumped in the normal direction with $v_j = 20 \text{ cm/s}$, $1\times$ higher than that on the smooth superhydrophobic surface ($v_j \approx 10 \text{ cm/s}$).

As R_1/R_2 decreased ($R_1/R_2 = 0.54 \pm 0.02$), droplet jumping with rotation was observed due to coalescence of suspended droplets on our microgrooved surface, as shown in Figure 5b. During coalescence, the generated surface counteracting force F from the liquid bridge impinging onto the plateau top deviated from the coalesced droplet center of mass, resulting in an angular momentum formed due to the generated moment, in turn enabling droplet clockwise rotation (see Section S4,

Supporting Information). This behavior is similar to the rotation of a rebounding–coalescing droplet on a superhydrophobic surface⁷² and the directional bouncing behavior of a droplet on a heterogeneous interface (seesaw effect).⁷³ Eventually, the rotational droplet jumped in a direction deviating from surface normal with an angle of $27 \pm 3^\circ$. With a further decrease of R_1/R_2 ($R_1/R_2 = 0.36 \pm 0.01$), the small suspended droplet was sucked into the large suspended droplet due to the Laplace pressure difference, and the momentum generated by liquid bridge impinging the top of the plateau was not enough to enable droplet detachment. Similarly, rotation of the coalesced droplet occurred due to the impinging position of the liquid bridge deviating from the center of mass of the droplet, as shown in Figure 5c. When the

radius of the small droplet was $\sim 250\ \mu\text{m}$, the critical radius ratio for coalescence-induced droplet jumping on smooth nanostructured superhydrophobic surfaces was reported to be approximately 0.49,³⁹ which was larger than the 0.36 observed on our microgrooved surface. Thus, the enhancement in surface-to-kinetic energy conversion efficiency from microgroove structures helps to decrease the critical radius ratio for coalescence-induced droplet jumping.

Figure 5e,f shows v_j and η as a function of R_1/R_2 for coalescence-induced jumping of suspended droplets on adjacent microgrooves and droplets on a smooth superhydrophobic surface. In order to eliminate the effect of droplet size on jumping velocity and energy conversion efficiency, the sizes of coalescing droplets on the smooth superhydrophobic surface were identical with that on our microgrooved surface for the same R_1/R_2 . When $R_1/R_2 = 1$, more surface energy is converted into kinetic energy for droplet jumping; hence, v_j and η of suspended droplets improved, consistent with traditional coalescence-induced droplet jumping on a smooth superhydrophobic surface. Furthermore, the suspended droplets possessed smaller surface adhesion, and the protruding ridge plateau between droplets redirected momentum, resulting in a further enhancement in v_j and η . Thus, the maximum v_j and η of suspended droplets could be achieved with $v_j = 20\ \text{cm/s}$ ($\approx 0.46U$) and $\eta \approx 18\%$, when $R_1/R_2 = 0.97 \pm 0.03$. The energy conversion efficiency is $2.18\times$ higher than droplet jumping on smooth superhydrophobic surfaces having similar R_1/R_2 ($\eta \approx 5.65\%$), confirming that the protruding structure between coalescing droplets can effectively enhance the coalescence-induced droplet jumping. Figure 5g shows the deviation angle of droplet jumping ($\Delta\theta$) as a function of R_1/R_2 . For coalescence-induced jumping of suspended droplets on adjacent microgrooves, $\Delta\theta$ increased with decreasing R_1/R_2 due to the rotation of coalesced droplets. The maximum $\Delta\theta$ for coalescence-induced jumping of suspended droplets was $\Delta\theta = 37 \pm 3^\circ$ on our microgrooved surface, when $R_1/R_2 = 0.5 \pm 0.02$.

Coalescence-Induced Jumping of Droplets on Adjacent Ridge Plateaus. The coalescence-induced jumping of droplets on adjacent ridge plateaus was also studied on our microgrooved surface. Figure 6a–d shows time-lapse images of coalescence-induced jumping of droplets on adjacent ridge plateaus with different radii ratios. As shown in Figure 6a, when $R_1/R_2 = 0.39 \pm 0.01$, the coalesced droplet was unable to jump due to the lack of surface reaction force. During coalescence, the small droplet detached from the top of the plateau and was sucked into the large droplet. Interestingly, droplet jetting was observed, similar to the partial coalescence of liquid droplets. The pinch-off of the liquid bridge is attributed to the competition between the axial and radial collapse inside the liquid bridge. When the axial collapse is sufficiently delayed by converging capillary waves, the radial collapse results in pinch-off and satellite droplet production.⁷⁴ At higher R_1/R_2 , the interaction between the coalesced droplet and ridge plateau increases. Figure 6b shows time-lapse images of coalescence-induced jumping of droplets on adjacent ridge plateaus with a higher $R_1/R_2 = 0.48 \pm 0.02$. During coalescence, the liquid bridge impinged on the top and edge of the ridge plateaus, which facilitated the coalesced droplet jumping in the normal direction with $v_j = 4.5\ \text{cm/s}$. As shown in Figure 6c, when R_1/R_2 increased to 0.74 ± 0.02 , the liquid bridge mainly impinged on an edge of the microgroove during coalescence, generating in-plane momentum. Thus, the

coalesced droplet jumped in a direction deviating from the surface normal with an angle of $\Delta\theta = 26 \pm 3^\circ$, and $v_j = 8.7\ \text{cm/s}$. Figure 6d shows that the liquid bridge forms above the microgroove, and the center of mass of the coalesced droplet is at the center line of the microgroove when $R_1/R_2 = 0.98 \pm 0.03$. During coalescence, the coalesced droplet detached first from the top of two ridge plateaus and then symmetrically impinged on both edges of the microgroove. Due to the presence of the microgroove, the coalesced droplet had sufficient deformation in the vertical direction and did not impinge on the bottom of the microgroove. As a result, the weakened impinging between the liquid bridge and surface resulted in the decreased v_j and η .

Figure 6e–g shows v_j , η , and $\Delta\theta$ as functions of R_1/R_2 , respectively, for coalescence-induced jumping of droplets on adjacent ridge plateaus and a smooth superhydrophobic surface. At the same R_1/R_2 , the sizes of coalescing droplets on the smooth superhydrophobic surface were identical with that on our microgrooved surface in order to eliminate the effect of droplet size. For traditional coalescence-induced droplet jumping on the smooth superhydrophobic surface, v_j and η increased with increasing R_1/R_2 . However, for coalescence-induced jumping of droplets on adjacent ridge plateaus, v_j and η increased at first with increasing R_1/R_2 and then decreased due to the changing of interaction between the coalesced droplet and microgroove structure. For $R_1/R_2 < 0.65$, v_j and η of droplets on adjacent plateaus were larger than those on the smooth superhydrophobic surface, due to the effect of plateau structures. However, v_j and η of droplets on adjacent plateaus became smaller than that on the smooth superhydrophobic surface at $R_1/R_2 > 0.65$, due to the weakened impinging of the formed liquid bridge during coalescence. When $R_1/R_2 = 0.98 \pm 0.03$, the coalescence-induced jumping velocity ($v_j = 5.5\ \text{cm/s}$) and energy conversion efficiency ($\eta \approx 1.4\%$) of droplets on adjacent ridge plateaus were 50% and 75% lower than those of the traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces ($v_j = 11\ \text{cm/s}$ and $\eta \approx 5.65\%$). Figure 6g shows $\Delta\theta$ as a function of R_1/R_2 . For coalescence-induced jumping of droplets on adjacent ridge plateaus, the deviation angle of droplet jumping increased at first with increasing R_1/R_2 and then decreased. At $R_1/R_2 < 0.74$, the coalesced droplet impinged on an edge of microgrooves to generated an in-plane momentum, resulting in the increase of $\Delta\theta$. For $R_1/R_2 > 0.74$, the coalesced droplet began to simultaneously impinge on both edges of the microgroove, resulting in reduced in-plane momentum, in turn decreasing the deviation angle of droplet jumping. The maximum $\Delta\theta$ of coalescence-induced jumping of droplets on adjacent ridge plateaus was $26 \pm 3^\circ$ on our microgrooved surface when $R_1/R_2 = 0.74 \pm 0.03$.

Discussion. Our work presents a study of coalescence-induced droplet jumping on hierarchical microgrooved superhydrophobic surfaces. The observed self-jumping of deformed droplets residing in microgrooves occurred due to the driving of Laplace pressure difference caused by the confinement. The observed behavior can reduce the accumulation of condensate in microgrooves and delay flooding of the condensing surface. Furthermore, the high surface-to-kinetic energy conversion efficiency of coalescence-induced jumping between deformed and undeformed droplets enables condensed droplet jumping at high surface subcooling.^{66,67} By rationally designing confining microstructures on hierarchical structured surfaces,

the combination of self-jumping of deformed droplets and coalescence-induced droplet jumping has the potential to facilitate the directional movement and rapid removal of condensate droplets, improving the refresh frequency of condensation, in turn enabling enhanced anti-icing performance and sustaining dropwise condensation.

The results (Figure 3) show that the larger the degree of droplets deformation within microgrooves, the larger generated Laplace pressure difference and surface energy stored in deformed droplets, in turn improving the self-jumping velocity and surface-to-kinetic energy conversion efficiency. We hypothesize that the aspect ratio of microgrooves has a significant effect on the self-jumping of deformed droplets within microgrooves and coalescence-induced jumping between deformed and undeformed droplets. The aspect ratio is usually defined as the ratio between width and height of the microgroove. We predict that decreased microgroove aspect ratio helps to enhance self-jumping of deformed droplets and coalescence-induced jumping between deformed and undeformed droplets, due to increased droplet deformation inside microgrooves induced by stronger confinement effects. Conversely, increased microgroove aspect ratios reduce droplet deformation, in turn weakening self-jumping of deformed droplets and coalescence-induced jumping between deformed and undeformed droplets. Thus, it would be interesting to further study the mechanism of self-jumping to facilitate self-cleaning of surfaces by considering the effects of liquid properties, surface wettability, geometry (V-shape,^{57,75} trapezoid,⁷⁶ or semicircle), and characteristic parameters (aspect ratio) of the microgroove structures in the future. As we know, the microgroove structures can result in the anisotropic wettability on the surface;⁶⁷ it would be worth further exploring the effect of anisotropic wettability of microgroove surfaces on coalescence-induced droplet jumping in our future study. Furthermore, we demonstrate that initial morphologies and positions of coalescing droplets play a key role on the coalescence hydrodynamics, resulting in high droplet jumping velocity ($\approx 0.74U$) and energy conversion efficiency ($\approx 46\%$). Due to the deformed droplet possessing more surface energy and larger Laplace pressure difference compared with the spherical droplet having an identical volume, it would be interesting to study the coalescence-induced jumping between two deformed droplets for further improved droplet jumping velocity and energy conversion efficiency. We can rationally design well-ordered hydrophilic defects at the bottom of microgrooves to steadily form deformed droplets residing in microgrooves during condensation, which facilitates the coalescence-induced jumping between deformed droplets for enhanced jumping droplet condensation. In addition to spatially engineering nucleation sites, the utilization of thermal conductivity gradients and heterogeneity opens up an avenue of future research to further utilize the confined jumping enhancement with dropwise condensation.^{77,78} By engineering the base of the ridges to be highly thermally conducting, and making the ridges from a superhydrophobic low-thermal-conductivity material such as a polymer, rational control of nucleation at the ridge base can be achieved during condensation.

As shown in Figure 5a, protruding ridge plateaus between droplets alter the momentum distribution inside coalescing droplets^{55,57} and intervene early during coalescence,⁷¹ resulting in enhanced jumping of suspended droplets. However, the microgroove between droplets on adjacent ridge plateaus

weakens recoil momentum of the impinging liquid bridge during coalescence, which leads to decreased droplet jumping velocity and energy conversion efficiency (Figure 6d). Thus, suppression of droplet formation on ridge tops and enhanced coalescence-induced jumping of suspended droplets through minimization of top dimensions of microstructures on hierarchical structured surfaces are needed. In addition, the microgroove structures change the interaction between the coalesced droplet and the surface, resulting in the droplet jumping direction deviating the surface normal, as shown in Figures 4b, 5b, and 6c. Therefore, future opportunities exist to develop structures that are capable of guiding the direction of droplet jumping by adjusting geometry and characteristic parameters of microstructures to facilitate high-efficiency water shedding. Indeed, the present work presents an avenue for further work on holistic design of droplet-scale structures to guide and define coalescence hydrodynamics for not only droplet jumping but also passive droplet motion on and from surfaces. This begs the question, what is the upper limit of jumping velocity and surface-to-kinetic energy conversion efficiency possible with the aid of symmetry breaking features, and can we achieve this ideal design in future surface structure design iterations?

CONCLUSIONS

In summary, coalescence-induced droplet jumping was studied on hierarchical microgrooved and smooth nanostructured superhydrophobic surfaces to explore the role of macrostructure and confinement on droplet jumping hydrodynamics. Various droplet jumping modes were observed on the hierarchical microgrooved superhydrophobic surface, including self-jumping of deformed droplets inside microgrooves, coalescence-induced jumping between deformed and undeformed droplets, coalescence-induced jumping of suspended droplets on adjacent microgrooves, and droplets on adjacent ridge plateaus. The droplet jumping velocity, direction, and energy conversion efficiency of each jumping mode were quantified and analyzed. The results indicate that confined microgroove structures and droplet deformation play key factors on improved droplet jumping velocity and energy conversion efficiency. The self-jumping of the deformed droplet in microgrooves is enabled with a Laplace pressure difference on the hierarchical microgrooved superhydrophobic surface. High droplet jumping velocity ($\approx 0.74U$) and energy conversion efficiency ($\eta \approx 46\%$) were achieved during the coalescence between deformed and undeformed droplets, which were $1.93\times$ and $6.67\times$ higher than traditional coalescence-induced droplet jumping on smooth superhydrophobic surfaces. Our work confirms that the protruding ridge plateau structures between coalescing droplets facilitate the enhancement in jumping velocity and energy conversion efficiency. The coalescence-induced jumping velocity ($\approx 0.46U$) and energy conversion efficiency ($\eta \approx 18\%$) were achieved during the coalescence between suspended droplets, which were $1\times$ and $2.18\times$ higher than coalescence-induced droplet jumping on smooth superhydrophobic surfaces. However, the coalescence-induced jumping velocity and energy conversion efficiency of droplets on adjacent ridge plateaus were 50% and 75% lower than coalescence-induced droplet jumping on smooth superhydrophobic surfaces due to weakened coupling between the symmetry breaking surface and liquid bridge. Our work presents an avenue for rational microstructure design to control coalescence hydrodynamics of

an adjacent droplet capable of enhancing the efficiency of droplet jumping and water shedding.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.0c01494>.

Calculation for the surface area and volume of deformed droplets residing in the microgrooves, determination of energy conversion efficiency for different coalescence-induced droplet jumping modes, force analysis of droplet jumping with rotation, error analysis, and descriptions of Videos S1–S13 (PDF)

Videos S1–S13 (ZIP)

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Author Contributions

Q.P., X.Y., L.J., and N.M. conceived the initial idea for this research. N.M. and L.J. guided the work. Q.P., Y.D., and C.D. manufactured the test samples. Q.P., X.Y., Y.D., and H.C. constructed the droplet jumping experiments. Q.P., X.Y., J.L., and L.L. analyzed the data. Q.P., X.Y., L.J., and N.M. edited and reviewed the manuscript.

Notes

The authors declare no competing financial interest.

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